MIL-T-5420B
6 April 1970
SUPERSEDING
MIL-T-5420A
7 MARCH 1951

MILITARY SPECIFICATION

TUBES, PITOT STATIC, ELECTRICALLY HEATED, AIRCRAFT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers two types of two-wire electrically heated pitot static tubes identified as follows:

Type I	Rated Voltage - 24 volt, direct current (DC)	
	AN5814-2	"L" shaped
	AN5815-2	Straight
Type II	Rated Voltage - 115 volt 400 Hz, single-phase alternating current (AC)	
	AN5814-3	"L" shaped
	AN5815-3	Straight

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein

Specifications

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawing, Engineering and Associated Lists
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

FSC 6610

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Standards

Federal

FED-STD-151	Metal, Test Methods

Military

MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1247	Marking, Functions and Hazard Designations of Hose, Pipe, and Tube Lines for Aircraft, Missile, and Space Systems
MS33586	Metal, Definition of Dissimilar
MS35207	Screw, Machine-Pan Head, Cross-Recessed, Carbon Steel, Cadmium Plated, UNF-2A
MS35338	Washer, Lock-Spring, Helical, Regular (Medium) Series

Air Force-Navy Aeronautical

AN3115	Receptacle-Airspeed Tube Electrical
AN5814	Tube, Pitot Static, L shaped, Electrically Heated
AN5816	Tube, Pitot Static, Straight, Electrically Heated

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. This specification makes provision for first article inspection.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials which are not specifically designated shall be of the best commercial quality, of the lightest practicable weight, and suitable for the purpose intended.

*3.3.1 Fungus-Proof materials. Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity.

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3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the tube, except where magnetic materials are essential.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.3.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4 Design and construction. The design of the tube shall conform to AN5814 or AN5816 as specified. The tubes shall be so constructed that no parts will work loose in service. They shall be built to withstand the normal strains of jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service without failure.

3.4.1 Drainage. The tube shall be so designed that when ice on the tube is melted under flight conditions, the water resulting therefrom shall drain from the tube and not be conducted into the impact pressure and static pressure lines where it may freeze. A drain shall also be provided for the removal of water from the mounting strut.

3.4.2 Electrical circuit. The electrical circuit of the tubes shall be of the two-wire type and shall be provided with a snap type receptacle conforming to AN3115-1.

*3.4.2.1 Heating element. The heating element placed in the nose of the [illegible] shall have a power rating of 145 watts at rated voltage with the atmosphere surrounding the tube at an ambient temperature of 25°C. Heating element current shall not exceed 15 amperes for type I tubes or 8 amperes for type II tubes under any operating condition specified herein.

3.4.3 Pitot and static connections. Pitot and static connections shall be as shown on AN5814 or AN5816.

3.5 Performance. The tubes shall satisfy the performance requirements of the tests specified in section 4.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto.

*3.7 Weight. The weight of the complete tube, without the AN3115 electrical assembly and mounting screws, shall not exceed 1.0 pound.

3.8 Finish. Protective coatings and finishes which will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used. The outside of the tube shall have a durable dull finish acceptable to the procuring activity. Aluminum alloy parts shall be covered

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with an anodic film conforming to MIL-A-8625.

3.9 Screw threads. The threads of all machine screws 0.060 inch diameter or larger shall conform to MIL-S-7742.

3.10 Identification of product. The following information shall be etched, engraved, or stamped on each tube in a location that will not interfere with proper operation of the tube

Name of Type (as shown on the AN standard)
Rated Voltage
AN Part No
Manufacturer's Part No
Contract or Order No
Manufacturer's Name or Federal Identification Code
U S Property

3.10.1 Other markings. Pitot and static lines shall be identified by the letters P and S, respectively, stamped, etched, or engraved on the lines as shown on AN5811 or AN5816. The pressure and static lines shall be color marked in accordance with MIL-STD-12[illegible]. The top of AN5816 tubes shall be stamped, etched, or engraved with the word "TOP", in letters 1/8 inch high, forward of the mounting shoulder

3.11 Protective caps. The total pressure opening and the threads of the pitot and static lines shall be protected by slip-on caps. The design of all protective devices shall be subject to the approval of the procuring activity.

3.12 Installation.

3.12.1 Installation instructions. Unless otherwise specified, the contractor shall furnish (with each tube) one printed copy of instructions for installation of the tube. Prior to printing, two copies shall be furnished to the procuring activity for approval.

3.12.2 Mounting screws. Four MS35207-22[illegible] screws and four MS35338-[illegible] lockwashers shall be furnished with each AN5816 tube. Four MS[illegible]-207-2[illegible] screws and four MS35338-[illegible] lockwashers shall be furnished with each AN5811 tube

3.12.3 Envelope. An envelope containing the installation instructions (when required) mounting screws and lockwashers shall be furnished with each tube. The following information shall be printed on the face of the envelope:

"IMPORTANT
This envelope contains installation instructions and mounting screws."

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3.13 Workmanship. The tube shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

*(a)	First article inspection	(see 4.4)
(b)	Quality conformance inspection	(see 4.5)

*4.3 Test conditions

*4.3.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches hg, a temperature of 25°C ± 5°C, and a relative humidity of 80% or less. Where tests are made with atmospheric pressure or temperature substantially different from the above value, proper allowance shall be made for the difference from the specified condition.

*4.4 First article inspection

4.4.1 Test samples. First article test samples shall consist of three tubes of each type being procured. Samples shall be representative of production items. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Test reports. When the contractor conducts the first article tests, he shall prepare a test report in accordance with MIL-STD-831. Three copies shall be furnished to the procuring activity.

4.4.3 First article tests. First article tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of individual tests and sampling tests.

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4.5.1 Individual tests. Each tube shall be subjected to the following tests as described under 4.6.

(a) Examination of Product

(b) Leakage

(c) Dielectric strength

(d) Electrical circuit

4.5.2 Sampling tests. Five tubes, selected at random from each 500 or fraction thereof produced, shall be subjected to the following tests as described under 4.6:

(a) Individual tests

(b) Vibration

(c) Power consumption

(d) Endurance

(e) Heat conductivity

(f) Magnetic effect

4.5.2.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.5.2.2 Individual tests may continue. For production reasons individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The tube shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, markings, and dimensions.

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4.6.2 Leakage. A pressure of 10 inches Hg shall be applied to each of the pressure lines with the total and static pressure orifices and drain holes sealed. The pressure source shall be sealed off at a point not more than two inches from the tube connectors. After one minute, the pressure shall not be below 9.95 inches Hg.

4.6.3 Dielectric strength. The following voltages, at a commercial frequency, shall be applied between each terminal and the case for a period of one minute.

Tube Type	Sea Level	50,000 Feet Altitude
I	550 v. rms	110 v. rms
II	1250 v. rms	500 v. rms

No insulation breakdown shall result

4.6.4 Electrical circuit. At an ambient temperature between 20° and 30° C (68° to 86°F), rated voltage shall be applied across the terminals of the tube for a period of two minutes. The power shall then be measured and shall be 145 ± 10 watts.

4.6.5 Vibration. The tube shall be mounted in a sleeve at the aft end and subjected to vertical and horizontal vibrations perpendicular to the tube centerline such that a point on the tip of the tube will oscillate through an amplitude of 1/4 inch. The test shall consist of three 1-hour periods at 1,000, 2,000, and 3,000 cpm. The type I tube shall be operated on 30 volts DC and the type II tube shall be operated on 120 volts, 400 Hz, AC continuously during this test. No leaks or failures shall result from this test.

4.6.6 Power consumption. The tube shall be supported vertically, immersed to within 1/4 inch of the mounting surface, in 1/2 gallon minimum volume of a water-ice mixture at an approximate temperature of 0°C (32°F) for a period of 10 minutes. Rated voltage shall then be applied across the terminals of the tube through a fuse (15-amp for type I tubes or 2-amp for type II tubes). The initial current surge shall not cause the fuse to fail.

4.6.7 Endurance. Type I tubes shall be operated on 30 volts DC and type II tubes shall be operated on 120 volt, 400 Hz, AC continuously for five hours. No damage which will affect corrosion resistance or proper operation of the tube shall result. This test may be combined with the vibration test by extending the period of power application to five hours.

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4.6.8 Heat conductivity. The tube shall be tested in both 3 liter and 5 liter volumes of water. The required volume of tap water (at or near room temperature) shall be placed in a glass beaker. The heated portion of the tube shall be supported vertically in the water to a depth of 1 1/2 inches for the 3 liter volume and 1/4 inch under the mounting surface for the 5 liter volume. The temperature of the water shall be measured and rated voltage shall then be applied to the tube for a period of 15 minutes. The water temperature shall again be measured and shall have risen not less than 5°C (9°F). The water shall be gently stirred during the test to assure uniform temperature.

4.6.9 Magnetic effect. The tube shall be held in various positions on a magnetic east-west line and five inches from the center of a short bar magnetic compass. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The test shall be made first with no voltage applied to the tube, then shall be repeated with the tube operating at a rated voltage. The deflection of the compass shall not exceed five degrees.

4.6.10 Airspeed and static pressure error.

4.6.10.1 Error at zero angle of attack. The tube shall be mounted in a wind tunnel and tested for both total and static pressures at the airspeed specified in Table I. Errors shall be determined by comparison with the results obtained under similar conditions with a pitot static tube of known accuracy and calibration. The error of the tube in indicated airspeed shall not exceed two knots and the static pressure errors shall be within the tolerances specified in Table I.

4.6.10.2 Errors at various angles of attack. The angle of attack shall be varied in 2° intervals from +12 to -10°. The error in indicated airspeed shall not differ from the error at zero angle of attack by more than two knots, and the error in static pressure shall not differ from the static pressure error at zero angle of attack by more than 0.15 inch of water.

TABLE I
Permissible Errors in Static Pressure

Indicated Air Speed	Tolerance
Knots	Inches of Water
50	0.05
75	0.10
100	0.15
125	0.20
150	0.25
175	0.30
200	0.35
225	0.40
250	0.45

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4.6.11 Salt spray. The tube, with electrical receptacle attached, shall be exposed to salt spray in accordance with FED-STD-151 for a period of 50 hours. The tube shall then be subjected to the tests specified in 4.5.1. No excessive corrosion or damage to any part of the tube shall result.

4.6.12 De-icing. The tube shall be tested in an icing wind tunnel at an indicated tunnel speed of 100 knots and an ambient temperature of -15° C ±5°C (+5°F -0°F). The nose of the tube shall be coated with an ice cap approximately 1/4 inch thick. The applicable voltage listed below shall then be applied.

For Type I tubes - 22 volts DC
For Type II tubes - 105 volts, 400 Hz, AC

The time required to clear the ice cap shall not exceed two minutes.

4.6.13 Cold resistance. The tube shall be subjected to a temperature of -65°C (-85°F) for 48 hours. The tube shall then be subjected to the tests specified in 4.5.1. No evidence of damage to the tube shall result.

4.6.14 Inspection packaging. Inspection of the preservation, packaging, packing and marking shall be conducted in accordance with the quality assurance provisions of MIL-STD-129 to determine conformance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

*5.1 Packaging.

*5.1.1 Level A. Tubes shall be packaged in accordance with Method IC1 of MIL-P-116, without the use of contact preservative, one each in a container conforming to PPP-B-636, weather-resistant class.

*5.1.2 Level B. Tubes shall be provided sufficient protection against physical and mechanical damage to assure safe delivery without degradation of the inter reliability, from the supply source to the first receiving activity for immediate use.

*5.2 Packing.

*5.2.1 Level A. Tubes packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, weather resistant class. When specified (see 6.2), tubes shall be packed in containers conforming to PPP-B-601, overseas type. Insofar as practicable, exterior containers shall be of uniform shape and size and of minimum cube and tare consistent with the protection required.

*5.2.2 Level B. Tubes packaged as specified in 5.1.2 shall be packed in domestic class exterior containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 shall apply.

*5.2.3 Level C. Packing which affords adequate protection against damage

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during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the suppliers commercial practice, provided the latter meets the requirements of this level

*5.3 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129

6 NOTES

6.1 Intended use. Pitot static tubes covered by this specification are intended for use in aircraft in conjunction with airspeed indicators for temperatures as low as -65°C (-85°F) The tubes are also used as a source of static pressure for altimeters and climb indicators.

*6.2 Ordering data. Procurement documents should specify the following.

(a) Title, number and date of this specification

(b) The AN number of the tube desired

(c) Required levels of packaging and packing

(d) When PPP-B-601 shipping containers are required

(e) Point of delivery for first article test samples

*NOTE The margins of this specification are marked with an asterisk to indicate where changes from the previous issue were made This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations Buyers and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians
Army - AV
Navy - AS
Air Force - 71

Reviewers
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Preparing Activity
Air Force - 71

Project Number
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22 R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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